Gokul Krishna Balaji

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Education

IIITDM Kancheepuram Chennai, India
B.Tech. in Computer Science and Engineering (Artificial Intelligence) | ***CGPA:*** *9.3/10.0* *Expected May 2028*

Experience

Virtusa Chennai, India
AI Engineering Intern *May 2026 – Jul 2026*

- Built a multimodal document-QA system over **PDF, DOCX, and PPTX** files, enabling questions over text, tables, and images using embeddings, table reasoning, and GPT-4o-based visual captions.
- Cut mean latency by ∼**48%** using lazy-loaded spaCy/Presidio models and document- and role-aware prompt caching, reaching **99.35%** cached prompt tokens.
- Developed reusable RAG evaluation scripts comparing **BM25, dense, and hybrid RRF** retrieval across **8+ metrics**; dense semantic search improved **recall@3 by ∼70%** and **NDCG@3 by ∼65%** over BM25-only search.
- Designed a multi-agent research assistant with **3 LangGraph agents** and **7 retrieval tools** spanning local files, Google Drive, and Tavily web search, using structured handoffs to generate source-grounded research reports.

Promptli | **Website** Remote
Software Developer, Part-time *Feb 2026 – July 2026*

- Built and deployed a production AI receptionist used by **4 businesses**, integrating LiveKit Agents, Twilio SIP, and **7 in-call tools** into a real-time voice workflow for scheduling, knowledge search, call notes, and follow-ups.
- Developed a React/Vite dashboard and Convex backend for customers, conversations, bookings, follow-ups, and agent management; used indexed, time-bounded queries to reduce calendar records scanned per request from **1,000 to 20** and booking records scanned from **1,000 to 1**.

HPRCSE Lab, IIITDM Kancheepuram Chennai, India
AI Research Intern *Dec 2025 – Feb 2026*

- Built an end-to-end PyTorch and Hugging Face training pipeline to fine-tune Mistral-7B for programming education and exposed the model through a FastAPI REST service with health and configurable generation endpoints.

Projects

Sagar Prahari | *React Native, Expo, TypeScript, Convex, GPS* Aug 2026

- Represented IIITDM Kancheepuram in contributing to the **Sagar Prahari** mobile app for **Swachh Sagar, Surakshit Sagar 2026**, a national coastal campaign by the **Ministry of Earth Sciences, Government of India.**
- Implemented citizen participation and event workflows involving photo-based proof, location data, event registration, and participation tracking.

InventoryForge | GitHub | *Java, Spring Boot, PostgreSQL, React, Docker, GitHub Actions* May 2026

- Designed a warehouse platform with **58 REST endpoints**, inventory and order state machines, RBAC, immutable ledgers, Flyway migrations, and Prometheus/Grafana monitoring.
- Implemented concurrency-safe allocation with PostgreSQL row-level locking and idempotent APIs; validated zero overselling across **50 concurrent order confirmations**.
- Built automated testing and CI with JUnit, jqwik, Testcontainers, Playwright, PIT, and GitHub Actions, achieving a **97% mutation score**.

ReFile | GitHub | *Python, FastAPI, Docker, LangChain, Supabase* Oct 2025

- Built a FastAPI backend converting natural-language media requests into validated Linux execution plans for video, audio, image, PDF, and OCR workflows.
- Sandboxed untrusted execution with Docker isolation, path sanitization, and ownership checks; won **2nd Place** at a national-level hackathon conducted by VIT Chennai.

Technical Skills

Languages: Java, Python, C++, TypeScript, JavaScript, SQL
Frameworks & Web: Spring Boot, FastAPI, Node.js, React, HTML, CSS, REST APIs, Convex
Databases & Systems: PostgreSQL, Supabase, relational databases, indexing, transactions, concurrency control, idempotency, state machines
Testing & DevOps: JUnit, jqwik, Testcontainers, Playwright, PIT, Git, Linux, Docker, GitHub Actions, Flyway, Prometheus, Grafana
Core CS: Data Structures and Algorithms, Object-Oriented Design, Database Management Systems, Complexity Analysis

Achievements & Leadership

- **Competitive Programming:** Solved **300+ problems** across LeetCode and HackerRank, covering dynamic programming, graphs, greedy algorithms, and complexity analysis.
- **Hackathons:** Won **1st place** at Innovation Unbound, a CodeChef FinTech Hackathon organized by **VIT Chennai**, **2 IIITDM hackathons**, and ranked **1st** in institute selection for the **Udbhav Inter-IIIT Hackathon**; placed **2nd** in the RetailTech track among 750+ teams at VOID:V1, VIT Chennai.
- **Leadership:** Served as a **Google Developer Student Clubs Coordinator**; helped organize Vashisht Hackathon 3.0 for **1,200+ teams** and delivered a hackathon-readiness session to **120+ attendees**.